

From Wall-Tap Delay to Operational Resolution Geometry

Origins, Uncertainty, Re-projection, and an Adversarial Battery for the Mature Claim

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Abstract

This paper gives a deliberately deflated but mathematically sharper account of the origin claim behind CTMT/RG. The childhood wall-tap observation is not treated as evidence for a new physical ontology. It is treated as the autobiographical origin of a precise operational question: what can be inferred when an observable event is confirmed only after mediation through unobserved degrees of freedom, with uncertainty present at every stage? The mature answer is formulated in terms of a resolved/null decomposition

$$O = R \oplus N,$$

positive-definite observation covariance, admissible morphisms, Schur-complement uncertainty correction, Mori–Zwanzig memory feedback, invariant coupling diagnostics, and transport of the resolved bundle. The central distinction is that statistical seepage, dynamical re-projection, and geometric holonomy are related but not identical. Statistical seepage is expressed by nonzero covariance coupling C_{RN} and the Schur complement. Dynamical re-projection requires the stronger bidirectional channel

$$R \xrightarrow{A_{NR}} N \xrightarrow{A_{RN}} R$$

and is represented by the memory kernel

$$K(\tau) = A_{RN} e^{A_{NN}\tau} A_{NR}.$$

Geometric mixing is the off-diagonal connection component $P_N(\nabla P_R)P_R$; closed-loop holonomy is a separate global test. We state an operational maturation definition, conditional necessity theorems, null-control soundness, and a minimal forced invariant set on a declared tested class. A Python appendix reproduces the attack batteries. The result is not a proof that nature must obey CTMT. It is a falsifiable operational geometry: under explicit admissibility assumptions, the invariant structure is constrained rather than arbitrary, and each accepted claim has a stated failure mode.

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1 Purpose and scope

This paper is about the mature form of an origin claim. It does not attempt to prove CTMT as physics. It attempts to make precise what survives when the original “wall-tap delay” intuition is translated into operational mathematics.

The intended claim is the following.

A wall-tap/delay/re-projection intuition matures into a partial-observability principle: when a system is observed only through a resolved sector R , coupled unresolved degrees of freedom N may leave structured, computable traces in R . These traces appear, depending on the formal layer, as Schur corrections, conditional covariance, memory kernels, estimation gains, transport-induced resolved/null mixing, or closed-loop holonomy. The words “seepage” and “re-projection” are admissible only as shorthand for these explicitly declared mechanisms.

The paper therefore separates three levels.

Layer I: mathematics. Fisher geometry under declared statistical assumptions, Schur complements, quotient observability, memory-kernel elimination, and bundle transport.

Layer II: operational inference. Resolved/null decomposition, recoverability, null-sector diagnostics, admissibility gates, and adversarial batteries.

Layer III: interpretation. The autobiographical wall-tap origin and possible physical applications. This layer motivates the questions but does not prove the mathematical statements.

Non-claim 1.1 (No physical completion). The present paper does not derive quantum mechanics, general relativity, turbulence, cosmology, or a new fundamental force. It establishes only conditional operational statements on declared model classes.

2 The origin observation: delay under uncertainty

The origin observation was simple. A child taps a wall; confirmation returns later. At the naive level this is only acoustics. It proves neither CTMT nor a hidden physical substrate. Its value is methodological: it exposes a primitive pattern of observation.

$$\text{action now} \longrightarrow \text{unobserved mediation} \longrightarrow \text{confirmation later.}$$

The crucial point is not merely that a delay $\Delta t = t_1 - t_0$ exists. A delay may be entirely contained in a resolved propagation model. The deeper point is that the observer does not access the full propagation path. The measured delay is a summary of an uncertain process involving medium, geometry, reflections, internal timing, and measurement threshold.

This can be written as an elementary inferential pattern:

$$Y = F(X_R, X_N) + \varepsilon,$$

where X_R denotes currently resolved variables, X_N denotes variables not directly resolved by the observer, and ε denotes measurement uncertainty and modelling error. The origin intuition becomes operational only after one asks when X_N leaves reproducible traces in the resolved data.

Remark 2.1 (What the wall-tap does and does not prove). The wall-tap observation motivates partial observability under uncertainty. It does not, by itself, imply null-sector coupling, memory kernels, or physical re-projection. Those require additional hypotheses and independent tests.

3 Observation geometry

Definition 3.1 (Resolution geometry). A local observation object is

$$\mathcal{G} = (O, g; R, N, \Sigma), \quad O = R \oplus N,$$

where O is a finite-dimensional observation space, g is a positive-definite observation metric, $R \subset O$ is the resolved sector, $N = R^\perp_g$ is the unresolved/null sector, and $\Sigma \succ 0$ is an observation covariance. Relative to the split, write

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{NR} & \Sigma_N \end{pmatrix}.$$

Definition 3.2 (Statistical seepage). Statistical seepage means nonzero resolved/null covariance coupling

$$C_{RN} \neq 0$$

whose effect on the resolved covariance is detected by the Schur complement

$$S_R = \Sigma_R - C_{RN} \Sigma_N^{-1} C_{NR}.$$

It is a statement about conditional covariance and estimation geometry, not by itself a statement about dynamical memory.

Definition 3.3 (Dynamical re-projection). For a linear resolved/null system

$$\dot{x}_R = A_{RR}x_R + A_{RN}x_N, \quad \dot{x}_N = A_{NR}x_R + A_{NN}x_N,$$

dynamical re-projection means that the composed channel

$$R \xrightarrow{A_{NR}} N \xrightarrow{A_{RN}} R$$

is nonzero and produces a memory kernel in the reduced resolved dynamics.

Definition 3.4 (Geometric mixing). Let P_R and $P_N = I - P_R$ be the projectors associated with a moving resolved/null split. The local off-diagonal transport component is

$$A_{RN}^\nabla = P_N(\nabla P_R)P_R.$$

This is connection data. Closed-loop holonomy requires additional loop-closure and curvature tests.

4 Admissibility assumptions

The word “necessary” is used only relative to the following assumptions.

Assumption 4.1 (A1: stable resolved rank). The resolved sector remains on a fixed rank stratum under admissible perturbations:

$$\text{rank } J_x = p$$

throughout the tested class, with declared tolerance away from rank walls.

Assumption 4.2 (A2: bounded conditioning). The Fisher or covariance objects used for inversion satisfy

$$\text{cond}(F_R) \leq \kappa_{\max}, \quad \text{cond}(\Sigma) \leq \kappa_{\max},$$

for a declared bound $\kappa_{\max}$.

Assumption 4.3 (A3: Fisher admissibility). Statistical morphisms are required to respect information monotonicity under admissible coarse-graining. Any use of Fisher geometry is conditional on the usual regularity and identifiability assumptions of the statistical model.

Assumption 4.4 (A4: quotient observability). Representations that produce indistinguishable observations under the declared observation map are identified. The classification is therefore on observational equivalence classes, not on arbitrary coordinate presentations.

Assumption 4.5 (A5: Schur-complement uncertainty control). When unresolved variables are eliminated in a Gaussian or covariance-based chart, the resolved uncertainty is evaluated using the Schur complement. A claim of statistical seepage requires a reproducible nonzero Schur correction.

Assumption 4.6 (A6: transport continuity and closure). Transport claims require continuous transport of the resolved projector. Holonomy claims require a closed base loop and closure of the transported resolution object within tolerance.

Assumption 4.7 (A7: bidirectional coupling for dynamical re-projection). A claim of dynamical re-projection requires a nonzero composed dynamic channel

$$A_{RN}e^{A_{NN}\tau}A_{NR}$$

for some $\tau \geq 0$. Covariance coupling alone is insufficient.

5 Operational maturation

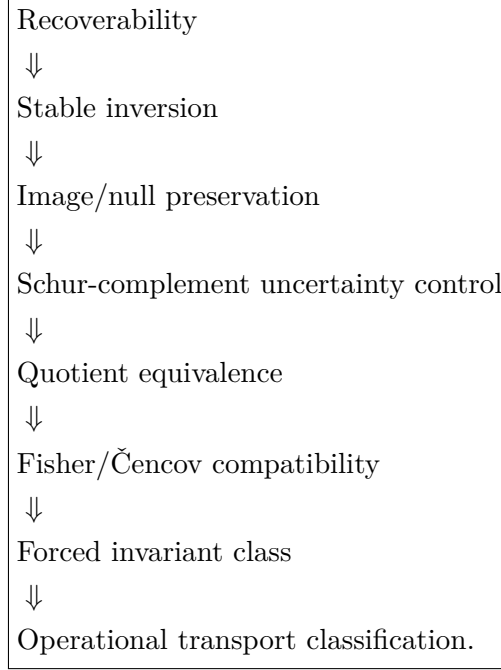
Definition 5.1 (Operational maturation on a model class). Let $\mathfrak{M}$ be a declared class of observation models with resolved/null decompositions, positive-definite covariance, admissible morphisms, and transport rules. A framework is operationally mature on $\mathfrak{M}$ if it supplies:

1. explicit admissibility hypotheses;

2. explicit falsification criteria;
3. invariant-based discrimination of admissible morphisms;
4. adversarial batteries whose failure modes are stated in advance;
5. and closure of every accepted class under all independent gates.

The term “mature” is therefore model-class-relative and operational, not an assertion of universal physical completeness.

The logical chain is:



This chain does not say that nature must obey CTMT. It says that once these operational assumptions are adopted, the invariant structure is constrained rather than arbitrary.

6 Theorems: what is forced, and under which hypotheses

Theorem 6.1 (Null-control soundness). *Let*

$$A = \begin{pmatrix} A_{RR} & A_{RN} \\ A_{NR} & A_{NN} \end{pmatrix}$$

generate a linear resolved/null system, and assume block-independent driving noise so that no statistical cross-noise is injected. If

$$A_{RN} = A_{NR} = 0,$$

then the stationary covariance is block diagonal,

$$C_{RN} = 0,$$

the Gaussian sector mutual information vanishes,

$$I(R; N) = 0,$$

the Schur complement reduces to

$$S_R = \Sigma_R,$$

and the re-projection memory kernel vanishes identically:

$$K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR} = 0.$$

Therefore the battery cannot infer resolved/null transport in a genuinely uncoupled block-independent system.

Proof. With block-diagonal dynamics and block-independent noise, the Lyapunov equation for the stationary covariance decouples sector by sector, hence the off-diagonal covariance block is zero. The Schur correction is quadratic in C_{RN} , so it vanishes. The Gaussian mutual information is determined by the canonical correlations of the whitened covariance coupling; these are zero. Finally, the memory kernel contains both off-diagonal dynamic blocks and is identically zero. $\square$

Theorem 6.2 (Elimination produces the re-projection kernel). *Consider*

$$\dot{x}_R = A_{RR}x_R + A_{RN}x_N, \quad \dot{x}_N = A_{NR}x_R + A_{NN}x_N,$$

with $x_N(0) = 0$. Eliminating x_N gives

$$\dot{x}_R(t) = A_{RR}x_R(t) + \int_0^t A_{RN}e^{A_{NN}(t-s)}A_{NR}x_R(s) ds.$$

Thus the unique reduced feedback operator generated by eliminating the unresolved sector is

$$K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}.$$

This is the precise dynamical meaning of re-projection.

Proof. Variation of constants gives

$$x_N(t) = e^{A_{NN}t}x_N(0) + \int_0^t e^{A_{NN}(t-s)}A_{NR}x_R(s) ds.$$

With $x_N(0) = 0$, substitute this expression into the equation for $\dot{x}_R$. The resulting convolution kernel is exactly $A_{RN}e^{A_{NN}\tau}A_{NR}$. $\square$

Warning 6.3 (Covariance does not imply dynamical re-projection). The implication

$$C_{RN} \neq 0 \implies K(\tau) \neq 0$$

is false. Covariance coupling may arise from one-way coupling or correlated driving noise. Therefore statistical seepage and dynamical re-projection must be stated separately.

Theorem 6.4 (Information equals estimation gain in the Gaussian chart). *In a calibrated linear-Gaussian resolved/null chart, let*

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{NR} & \Sigma_N \end{pmatrix}, \quad S_R = \Sigma_R - C_{RN}\Sigma_N^{-1}C_{NR}.$$

If ρ_i are the canonical correlations of the whitened coupling, then

$$\frac{1}{2} \log \frac{\det \Sigma_R}{\det S_R} = -\frac{1}{2} \sum_i \log(1 - \rho_i^2) = I(R; N).$$

Proof. Whiten Σ_R and Σ_N , so the coupling block has singular values ρ_i . Then

$$\det S_R = \det \Sigma_R \det(I - \tilde{C}\tilde{C}^T) = \det \Sigma_R \prod_i (1 - \rho_i^2).$$

Taking the logarithmic volume ratio gives the stated expression. The same expression is the mutual information of jointly Gaussian sectors. $\square$

Theorem 6.5 (Conditional operational necessity). *Assume A1–A7 on a declared model class $\mathfrak{M}$. Then the following diagnostics are forced by the admissibility gates and are not arbitrary decorations:*

$$F_R, \quad C_{RN}, \quad S_R, \quad K(\tau), \quad [P_R], \quad P_N(\nabla P_R)P_R, \quad \text{Hol}(\gamma)$$

whenever their corresponding hypotheses are invoked. In particular, Fisher diagnostics require stable rank and bounded conditioning, statistical seepage requires reproducible Schur correction, dynamical re-projection requires bidirectional dynamic coupling, and holonomy requires closed-loop transport closure.

Proof sketch. Fisher diagnostics are meaningful only on identifiable, bounded-condition strata. Stable rank prevents arbitrary changes of the resolved image without crossing a declared wall. Schur-complement uncertainty control is the covariance correction forced by eliminating Gaussian unresolved variables. The memory kernel is forced by variation of constants once bidirectional dynamic coupling is assumed. Quotient observability forces classification on equivalence classes rather than coordinate representations. Finally, transport continuity supplies local off-diagonal connection data, while holonomy is defined only for closed transported objects. Each diagnostic corresponds to one admissibility requirement whose omission produces a failure mode demonstrated in the attack battery. $\square$

7 Minimal forced invariant set on the tested class

For the tested operational class, use the invariant tuple

$$\mathcal{I} = \{I_F, I_N, I_W, I_S, I_Q\},$$

where:

- I_F = Fisher order / conditioning class,
- I_N = resolved/null and Schur-coupling class,
- I_W = winding or closed-loop transport class,
- I_S = self-reference or stability/recoverability class,
- I_Q = observational quotient class.

Theorem 7.1 (Minimal forced invariant set, tested-class version). *Let admissible morphisms on the tested class satisfy Fisher monotonicity, stable inversion, bounded conditioning, quotient observability, and transport continuity. Then every accepted morphism preserves*

$$\mathcal{I} = \{I_F, I_N, I_W, I_S, I_Q\}.$$

Moreover, on the adversarial classes tested in this paper, no proper subset of $\mathcal{I}$ distinguishes all admissible and violating cases.

Proof strategy. The first claim follows componentwise from the assumptions. The second is not a universal classification theorem; it is a tested-class minimality result. Dropping I_F loses rank-wall and ill-conditioning failures. Dropping I_N loses statistical resolved/null coupling. Dropping I_W loses closed-loop transport defects. Dropping I_S loses recovery/stability failures. Dropping I_Q overcounts observationally equivalent representations. The battery in Section 8 supplies witness classes for these losses. $\square$

Remark 7.2 (What uniqueness is not claimed). This theorem does not say $\mathcal{I}$ is the unique invariant basis for all possible mathematical structures. It says that within the declared operational framework and tested adversarial classes, each element blocks a distinct failure mode.

Invariant	Forced by	Failure if omitted
I_F Fisher order	statistical admissibility, conditioning	artificial info. gain, rank-wall blindness
I_N null/Schur class	unresolved-sector uncertainty control	coupling mistaken for noise
I_W winding/holonomy	transport continuity and closure	path-dependent transport missed
I_S stability/recoverability	inverse stability	unstable inverse accepted
I_Q quotient class	observational equivalence	indistinguishable charts overcounted

8 Adversarial battery

The battery is designed to decide which claims require softening. It tests five constructed cases:

1. uncoupled dynamics;
2. one-way $N \rightarrow R$ coupling;
3. one-way $R \rightarrow N$ coupling;
4. bidirectional coupling;
5. statistical correlation without dynamic coupling.

8.1 Necessity attack results

Case	A_{RN}	A_{NR}	$K \neq 0$	$C_{RN} \neq 0$	$I(R; N) > 0$	Schur shift
Uncoupled	no	no	no	no	no	no
One-way $N \rightarrow R$	yes	no	no	yes	yes	yes
One-way $R \rightarrow N$	no	yes	no	yes	yes	yes
Bidirectional	yes	yes	yes	yes	yes	yes
Statistical correlation only	no	no	no	yes	yes	yes

The key failure is:

$$C_{RN} \neq 0 \not\Rightarrow K(\tau) \neq 0, \quad I(R; N) > 0 \not\Rightarrow K(\tau) \neq 0.$$

Therefore nonzero covariance and sector mutual information do not prove dynamical re-projection. They prove statistical dependence only.

8.2 Computed battery values

Case	$\ C_{RN}\ $	MI	Schur shift	Kernel int.	cond
Uncoupled	0.000000	0.000000	0.000000	0.000000	1.333
One-way $N \rightarrow R$	0.033689	0.065107	0.009973	0.000000	2.038
One-way $R \rightarrow N$	0.028828	0.048073	0.006483	0.000000	1.864
Bidirectional	0.082515	0.274465	0.044895	0.388682	3.639
Statistical correlation only	0.035147	0.085601	0.010462	0.000000	2.138

8.3 Layer sweep: structured uncertainty

A second battery varies a single coupling knob α in an Ornstein–Uhlenbeck model. The whole machine reacts in lockstep only under bidirectional dynamic coupling.

α	$\ C_{RN}\ $	MI	Est. gain	identity error	kernel norm
0.00	0.0000	0.000000	0.000000	0.00e+00	0.0000
0.10	0.0071	0.002702	0.002702	1.17e-16	0.0043
0.25	0.0181	0.016995	0.016995	2.46e-16	0.0271
0.45	0.0338	0.056058	0.056058	1.87e-16	0.0877
0.70	0.0572	0.141013	0.141013	1.94e-16	0.2121
1.00	0.0965	0.310832	0.310832	1.11e-16	0.4329

Finding 8.1 (Re-projection reconstruction). For $\alpha = 0.70$, an impulse in R with $x_N(0) = 0$ excited the unresolved sector and returned via the memory kernel. The resolved-only no-memory model had RMSE 0.017571, while the memory-kernel reconstruction had RMSE 0.000766, an improvement factor of about 22.9. The corresponding null control with $\alpha = 0$ had hidden excitation 0, mutual information 0, and resolved-only RMSE at numerical roundoff.

8.4 Geometric attack

The geometric counterexample shows:

$$\|P_N(dP_R)P_R\| \approx 1, \quad \text{Hol}(\text{backtracking loop}) = 0.$$

Thus local mixing is not sufficient for holonomy. Holonomy requires a nontrivial closed-loop transport test.

8.5 Rank-wall attack

A Fisher matrix can remain formally defined while becoming operationally unusable:

$$\lambda_{\min}(F_{\text{good}}) = 0.04, \quad \lambda_{\min}(F_{\text{wall}}) = 10^{-18}, \quad \text{cond}(F_{\text{wall}}) \approx 10^{18}.$$

Thus Fisher admissibility must include stable rank and bounded conditioning.

9 Defendability verdict

Claim	Battery verdict	Final wording
Delay alone implies seepage	fails	delay motivates partial observability
$C_{RN} \neq 0$ implies re-projection	fails	implies statistical dependence only
$I(R; N) > 0$ implies memory feedback	fails	MI measures dependence in Gaussian chart
Bidirectional coupling gives memory kernel	survives	dynamic re-projection under A7
Null coupling makes diagnostics vanish	survives strongly	null-control soundness theorem
Local mixing implies holonomy	fails	holonomy requires closed-loop transport
Fisher admissibility preserves rank	fails	add stable rank and bounded conditioning
Invariants necessary for physics	fails	necessary under operational assumptions
Minimal invariant set on tested class	survives conditionally	tested-class minimality only

10 Scope and limits

The present results establish:

1. conditional necessity of the invariant class under declared assumptions;
2. falsifiability through null controls;
3. synthetic validation of statistical seepage and dynamic re-projection as separate layers;
4. stability of memory-kernel reconstruction in the tested class;
5. and transport discrimination only when closed-loop hypotheses are satisfied.

They do not establish:

1. universal physical uniqueness;
2. a new foundation of physics;
3. quantum foundations;
4. relativistic field equations;
5. turbulence or Navier–Stokes physics;
6. cosmological implications;
7. or completeness outside the declared model class.

11 Conclusion

The wall-tap origin does not prove CTMT. Its mature role is different and more defensible: it identifies the primitive operational situation in which a resolved event is confirmed only after uncertain unobserved mediation. Once this is translated into resolved/null mathematics, the claims become smaller and stronger.

The final structure is:

partial observability + structured uncertainty + declared coupling type + admissibility gates $\implies$ specific forced diagnostics.

But each diagnostic has its own gate:

C_{RN} requires statistical dependence,

$K(\tau)$ requires bidirectional dynamic coupling,

$\text{Hol}(\gamma)$ requires closed-loop transport closure,

Fisher classification requires stable rank and bounded conditioning.

Thus mature RG should not be presented as a universal physical theory in this paper. It should be presented as an operational geometry with explicit assumptions, forced invariants, null controls, and failure modes. This is not weaker than the origin story. It is the part of the origin story that can survive mathematical review.

A Python appendix: necessity attack battery

```

import numpy as np
from scipy.linalg import solve_continuous_lyapunov, expm
np.set_printoptions(precision=6, suppress=True)

r=q=2; n=4
A_RR=np.array([[ -1.2, 0.25], [ -0.1, -1.0]])
A_NN=np.array([[ -1.1, 0.2], [ -0.15, -1.3]])
B_RN=np.array([[ 0.7, -0.2], [ 0.15, 0.5]])
B_NR=np.array([[ 0.45, 0.25], [ -0.25, 0.4]])
Q=0.2*np.eye(n)

def stat_cov(A,Quse=Q):
    S=solve_continuous_lyapunov(A,-Quse)
    return (S+S.T)/2

def metrics(A,Quse=Q):
    S=stat_cov(A,Quse)
    SR=S[:,r,:]; SN=S[r,:r]; C=S[:,r:]
    Schur=SR-C@np.linalg.inv(SN)@C.T
    wr,Ur=np.linalg.eigh(SR); wn,Un=np.linalg.eigh(SN)
    M=Ur@np.diag(1/np.sqrt(wr))@Ur.T @ C @ Un@np.diag(1/np.sqrt(wn))@Un.T
    rhos=np.clip(np.linalg.svd(M,compute_uv=False),0,1-1e-12)
    MI=float(-0.5*np.sum(np.log(1-rhos**2)))
    grid=np.linspace(0,8,801)
    Kint=float(np.trapz([
        np.linalg.norm(A[:,r,r:]@expm(A_NN*t)@A[r,:r], 'fro')
        for t in grid], grid))
    return dict(
        stable=bool(np.max(np.real(np.linalg.eigvals(A)))<0),
        Cnorm=float(np.linalg.norm(C, 'fro')),
        MI=MI,
        SchurShift=float(np.linalg.norm(SR-Schur, 'fro')),
        Kint=Kint,
        cond=float(np.linalg.cond(S)),
        minEig=float(np.linalg.eigvalsh(S).min()))

cases={
'T0_uncoupled':np.block([[A_RR,0*B_RN],[0*B_NR,A_NN]]),
'T1_oneway_N_to_R':np.block([[A_RR,B_RN],[0*B_NR,A_NN]]),
'T2_oneway_R_to_N':np.block([[A_RR,0*B_RN],[B_NR,A_NN]]),
'T3_bidirectional':np.block([[A_RR,B_RN],[B_NR,A_NN]])
}

for name,A in cases.items():
    print(name, metrics(A))

# statistical correlation without dynamic coupling
Q_cross=Q.copy()
Q_cross[:,r,:]=np.array([[0.06,-0.02],[0.01,0.05]])
Q_cross[r,:r]=Q_cross[:,r:].T
A_block=cases['T0_uncoupled']
print('T4_statistical_correlation_only', metrics(A_block,Q_cross))

# geometric local-mixing but zero-holonomy backtracking loop
def frame(theta):
    u1=np.array([np.cos(theta),0,np.sin(theta),0])

```

```

u2=np.array([0,1,0,0])
return np.column_stack([u1,u2])

def proj(theta):
    F=frame(theta)
    return F@F.T

th=0.4; eps=1e-5
P=proj(th)
dP=(proj(th+eps)-proj(th-eps))/(2*eps)
local_mix=np.linalg.norm((np.eye(4)-P)@dP@P,'fro')
pts=list(np.linspace(0.4,0.7,100))+list(np.linspace(0.7,0.4,100))
Fs=[frame(x) for x in pts]
H=np.eye(2)
for i in range(len(Fs)-1):
    O=Fs[i+1].T@Fs[i]
    U,s,Vt=np.linalg.svd(O)
    H=(U@Vt)@H
hol_angle=abs(np.arctan2(H[1,0]-H[0,1],H[0,0]+H[1,1]))
print('local_mix',local_mix,'backtrack_holonomy',hol_angle)

# rank wall
J_good=np.diag([1.0,0.5,0.2])
J_wall=np.diag([1.0,0.5,1e-9])
F_good=J_good.T@J_good
F_wall=J_wall.T@J_wall
print('rank wall',np.linalg.eigvalsh(F_good).min(),
      np.linalg.eigvalsh(F_wall).min(),np.linalg.cond(F_wall))

```

B Python appendix: structured uncertainty and re-projection battery

```

import numpy as np
from scipy.linalg import solve_continuous_lyapunov, expm

r=q=2; n=4
A_RR=np.array([[ -1.30,0.25],[ -0.15,-0.95]])
A_NN=np.array([[ -1.10,0.35],[ -0.20,-1.25]])
A_RNO=np.array([[ 0.72,-0.25],[ 0.15,0.55]])
A_NRO=np.array([[ 0.48,0.18],[ -0.30,0.46]])
Q=0.22*np.eye(n)

def canonical_rhos(S):
    SR=S[:,r:]; SN=S[r:,r:]; C=S[:,r:]
    wr,Ur=np.linalg.eigh(SR)
    wn,Un=np.linalg.eigh(SN)
    SR_mh=Ur@np.diag(1/np.sqrt(wr))@Ur.T
    SN_mh=Un@np.diag(1/np.sqrt(wn))@Un.T
    return np.linalg.svd(SR_mh@C@SN_mh,compute_uv=False)

def mi_from_rhos(rhos):
    return float(-0.5*np.sum(np.log(1-rhos**2)))

for alpha in [0.0,0.10,0.25,0.45,0.70,1.00]:
    A=np.block([[A_RR,alpha*A_RNO],[alpha*A_NRO,A_NN]])
    S=solve_continuous_lyapunov(A,-Q); S=(S+S.T)/2

```

```

SR=S[:,r,:]; SN=S[r:,r,:]; C=S[:,r,:]
Schur=SR-C@np.linalg.inv(SN)@C.T
rhos=canonical_rhos(S)
MI=mi_from_rhos(rhos)
gain=0.5*np.log(np.linalg.det(SR)/np.linalg.det(Schur))
grid=np.linspace(0,10,1001)
Kint=np.trapz([
    np.linalg.norm((alpha*A_RN0)@expm(A_NN*t)@(alpha*A_NRO),'fro')
    for t in grid], grid)
print(alpha,np.linalg.norm(C,'fro'),MI,gain,abs(MI-gain),Kint)

# impulse re-projection at alpha=0.70
alpha=0.70
A=np.block([[A_RR,alpha*A_RN0],[alpha*A_NRO,A_NN]])
xR0=np.array([1.0,-0.4])
x0=np.r_[xR0,np.zeros(q)]
ts=np.linspace(0,6,601); h=ts[1]-ts[0]
X_full=np.array([expm(A*t)@x0 for t in ts])
R_full=X_full[:,r]
N_full=X_full[:,r:]
R_nomem=np.array([expm(A_RR*t)@xR0 for t in ts])
Kvals=np.array([(alpha*A_RN0)@expm(A_NN*t)@(alpha*A_NRO) for t in ts])
R_mem=np.zeros_like(R_full); R_mem[0]=xR0
for k in range(len(ts)-1):
    conv=sum(Kvals[j]@R_mem[k-j] for j in range(k+1))*h
    R_mem[k+1]=R_mem[k]+h*(A_RR@R_mem[k]+conv)
print('RMSE no memory',np.sqrt(np.mean((R_nomem-R_full)**2)))
print('RMSE memory',np.sqrt(np.mean((R_mem-R_full)**2)))
print('max hidden excitation',np.max(np.linalg.norm(N_full,axis=1)))

```

References

- [1] R. Zwanzig, “Memory effects in irreversible thermodynamics,” *Physical Review* 124 (1961), 983–992.
- [2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- [3] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Wiley, 1963–1969.